

SDLC Maintenance Phase: Quick Summary

The maintenance phase is a continuous cycle that begins immediately after deployment. It is essential for ensuring software remains operational, secure, and relevant over time.

Why It Never Ends

Software is a "living" product that requires constant updates due to:

- **Environmental Shifts:** Adjusting to new hardware, OS updates, or API changes.
- **Security Needs:** Patching vulnerabilities against emerging cyber threats.
- **User Evolution:** Adapting to changing business requirements and market demands.

3 Key Maintenance Activities

1. **Corrective:** Fixing bugs and errors reported by users.
2. **Adaptive:** Modifying the system to remain compatible with environmental changes.
3. **Perfective:** Enhancing existing features or performance based on user needs.

Tools for Management

- **Jira:** Used for issue tracking, backlog management, and sprint planning.
- **ServiceNow:** Centralizes incident tracking and change management workflows.

Influence of User Feedback

User feedback is the primary driver for prioritizing maintenance. It highlights usability friction, missing features, and critical bugs, allowing developers to focus efforts on the updates that provide the most value to the end user.

Case Study: Proactive vs. Reactive Maintenance

Organizations that prioritize regular, proactive patch management successfully prevent security breaches before they occur. Conversely, neglecting maintenance often leaves systems vulnerable to known exploits, which can turn minor security gaps into catastrophic system failures or massive data breaches.

References

- ADEVS. (2026, February 10). *Types of Software Maintenance*. <https://adevs.com/blog/types-of-software-maintenance/>
- IBM. (n.d.). *What is the Software Development Lifecycle (SDLC)?*. <https://www.ibm.com/think/topics/sdlc>
- MOCA. (n.d.). *Why User Feedback Matters for Website Maintenance*. <https://www.moca.co.nz/post/why-user-feedback-matters-for-website-maintenance>
- N-ix. (2025, August 18). *Key types of software maintenance*. <https://www.n-ix.com/types-of-software-maintenance/>
- Serdao. (2025, October 21). *How System Maintenance Improves Security*. <https://www.serdao.com/blog/how-system-maintenance-and-updates-improve-security-and-performance>